SCI-POINT

Scientific Rationale

Scientific contract v0.1   Document revision r0.4

Concise scientist-facing view

Authority status. The complete scientific authority is SCI-POINT-NORMATIVE-CORE/v0.1-r0.4. This view explains its conditional known-source, per-array Pointing contract without reproducing the complete formal registers. The compatibility, formal-error, and full-map-RMS methods and every exact numerical route remain unavailable pending separate owner approval. Implementation conformity has not been assessed.

Scientific owner	Grant Wilson
Status	Scientific authority frozen; implementation conformity unassessed
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Reader summary

Input. One exact observation-local, per-array MAP, JINC, FLT-FIXED, or FLT-MATCHED signal product; exact state and optional terminal FRUIT ancestry; a valid expected source position and AltAz tangent boundary; and explicit requested method/use state.

Conditional output. A six-role zero-background elliptical-Gaussian fit and signed source displacement $\boldsymbol{\Delta}_{\mathrm{POINT}} = \widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}_{\mathrm{expected}}$ in arcseconds, with exact identities, states, dependencies, uncertainty availability, diagnostics, and separately owned use decisions.

Present numerical status. Unavailable. The missing numerical authorities are not inferred.

The physical meaning in one sentence is: for one identified source and one exact processed map, fit the approved Gaussian family in the physical AltAz tangent metric, independently establish that the fitted feature is the source, and report its centroid minus the authoritative expected position while preserving every parent, method, response, uncertainty, and use limitation.

exact parent + source reference + AltAz tangent boundary ⟶ conditional fit + association

↓

signed displacement + bounded diagnostics + separate use decisions

Contents

1 Why the contract is conditional

A mature operation can have a stable scientific purpose while still lacking a content-bound numerical procedure. Choosing a familiar least-squares objective, a conventional Gaussian width, or an assumed inverse-variance weight would silently replace that operation. SCI-POINT therefore fixes the scientific estimand, six parameter roles, zero-background family, coordinate measurement, four parent families, ownership, typed state semantics, and falsifiable consequences while leaving unavailable details unavailable.

This separation lets the exact compatibility, formal-error, and full-map-RMS methods be recovered later without changing the primary estimand. It also prevents a missing optional diagnostic, response, or downstream decision from erasing an otherwise authorized independent fit fact.

The producer lifecycle, component-identifiability state, and named-use evaluation are independent. SCI-VAL retains four axes: request (`requested`, `not_requested`); applicability (`applicable`, `inapplicable`, `applicability_unknown`); eligibility (`eligible`, `ineligible`, `decision_unavailable`); and realization (`realized`, `incomplete`, `failed`, `not_produced`). Where the controlling crosswalk makes no applicability or eligibility proposition, no value exists; absence is not another token. A successfully written unknown-scope or conflict accounting artifact is `realized`; `failed` means production of the decision artifact failed.

2 Parent routes and ownership

SCI-POINT consumes exactly one observation-local per-array parent and never selects, substitutes, equates, or falls back among route families. The normative core binds the exact POINT-facing role identities; readable role labels are used below.

Family	POINT-facing signal role	Required distinction
MAP	Observation-level normalized signal	Exact observation-level normalized signal; never a coadd.
JINC	Normalized JINC signal	Exact signed normalized signal; normalization and response companions do not replace the signal.
FLT-FIXED	Fixed transformed signal	Exact transformed signal with parent, operator, normalization, response, edge/support, and covariance state.
FLT-MATCHED	Matched-template amplitude field	Exact template-amplitude field with template, noise/covariance selector, normalization, phase/origin, response, and support.

Terminal FRUIT is ancestry, not a fifth route. It retains the exact terminal MAP, JINC, FLT-FIXED, or FLT-MATCHED identity plus complete terminal iteration, generation, recurrence, restart, response, support, covariance, calibration, null/additive-reference, lifecycle, and provenance. Intermediate iterations and coadds are outside base v0.1.

The parent producer owns the parent estimand, units, calibration, response, support, covariance, lifecycle, and provenance. The source-reference producer owns the expected position. AST owns the physical AltAz tangent realization. POINT owns the per-array fit and the sign of its displacement measurement. Pointing support owns aggregation and correction construction; AST owns correction application. SCI-BEAM owns beam products; CAL/TolProj owns photometric-transfer admission; named QC processes own their thresholds and actions; NOI owns separately authorized empirical uncertainty; SCI-VAL registers and evaluates profiles but authors no scientific policy.

3 Model, measurement, search, and association

For physical tangent coordinate $\boldsymbol{u}_q$, the base model is

$$g_q(A, \boldsymbol{\mu}, \boldsymbol{\Sigma}_g) = A \exp\Big[-\tfrac{1}{2}(\boldsymbol{u}_q - \boldsymbol{\mu})^\mathsf{T} \boldsymbol{\Sigma}_g^{-1} (\boldsymbol{u}_q - \boldsymbol{\mu})\Big] . \tag{1}$$

The symmetric positive-definite shape has ordered, strictly positive principal extents. The six roles are amplitude, two centroid components, two principal widths, and orientation. No constant, gradient, local baseline, or second model family is admitted. At exact circularity the orientation is `undefined_by_model_symmetry`; no finite placeholder is a measurement.

The exact AST boundary maps parent coordinates into the declared physical AltAz tangent metric. The signed measurement is

$$\boldsymbol{\Delta}_{\text{POINT}} = \widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}_{\text{expected}}, \qquad [\boldsymbol{\Delta}_{\text{POINT}}] = \text{arcsec}. \tag{2}$$

Search center, expected position, selected seed, fitted centroid, and source association are distinct. Search and fallback locate candidates; they never define the astrometric zero or establish identity. Every central, global, restart, or retry branch requires an independent `POINT-SOURCE-ASSOCIATION-STATE=established` before displacement, source-associated amplitude, or an intended-source diagnostic can be created.

4 Method authorities and present availability

Authority	Present state	Enabled when available	Blocked while absent	Unaffected	Release condition
Compatibility method v0.1	unavailable; owner approval pending	Numerical fit and all fit-derived roles	Fit, components, displacement, diagnostics, formal errors	Request, parent, boundary and typed accounting	Exact immutable method, numerical parent, and route binding
Formal-error method v0.1	unavailable; owner approval pending	Marginal formal errors and signed formal standardization	Formal errors, unsupported joint covariance and dependent uses	Authorized fitted values and non-formal roles	Exact error method with conditioning and failure rules

Authority	Present state	Enabled when available	Blocked while absent	Unaffected	Release condition
Full-map-RMS method v0.1	unavailable; owner approval pending	RMS scale and signed amplitude/RMS diagnostic	RMS scale, diagnostic and approved aliases	Fit, displacement and formal errors	Exact population, centering, weighting, support, denominator, and lineage rules

The conditional estimator is only

$$\widehat{\boldsymbol{\theta}} = \mathrm{Resolve}_{\boldsymbol{\theta}\in\Theta_{\mathrm{effective}}}\, D(m, g_{\boldsymbol{\theta}}; W_{\mathrm{fit}})\,. \tag{3}$$

The exact objective, rows, constraints, comparison, convergence, retry, and solution rule remain gated. W_{fit} is the exact consumed object; uniform, reliability, covariance, inverse-covariance, or other approved source roles are not aliases. Every mapping binds identity, source/target units, normalization, support, state, claim ceiling, and provenance.

Every exact route instance is also unavailable pending its upstream product role, contract version, source digest, owner approval, compatibility state, generation, and parent-response identity/status. The four draft named-use profiles remain unregistered and unevaluable pending owner review. Separate authority is required for fixed response, POINT-full-procedure/parent-fixed response, whole-chain response, observational bias/accuracy, and any requested covariance representation.

5 Products, dependencies, diagnostics, and edge behavior

The consolidated table states minimum scientific dependencies; the normative core controls every exact field and terminal rule.

Role	Scientific quantity	Minimum dependencies	Unavailable result	Claim ceiling
Fit result	Six-role conditional fit	Compatibility method, exact parent/route, support, plan and fit realization	No numerical fit	Not association, beam, catalog or correction
Fit-amplitude component	Signed fitted A	Conforming fit and component state	Component status retained	Parent-unit fit component; not source attribution

Role	Scientific quantity	Minimum dependencies	Unavailable result	Claim ceiling
Displacement measurement	$\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}_{\text{expected}}$ in AltAz	Identifiable centroid, expected position, exact basis, established association	Displacement unavailable; fit may remain	Measurement, never correction
Source-associated amplitude	Fitted A attributed to intended source	Fit amplitude plus established association	Source-associated role unavailable	Candidate input only; not universal flux
Signed derived diagnostics	$\widehat{A}/\text{RMS}_{\text{full}}$; $\widehat{A}/\sigma_A$	Separate request, exact amplitude, corresponding method, positive finite denominator, claim-appropriate association	Only dependent diagnostic unavailable	Descriptive ratio or formal standardization; not significance
Effective-shape diagnostic	Processed-map widths and angle	Conforming fit and component states	Per-role status retained	Not intrinsic beam or unique cause
Fit QC metrics	Fit components and states for named QC	Immutable fit plus owner profile/evaluation	QC decision unavailable	No automatic threshold, cause or action
Association state	Established, unavailable, or failed	Independent method, domain, cause and branch coverage	Source-attributed roles blocked	Never implied by peak or finite fit
Response, bias, covariance states	Exact separately scoped companions	Exact authority and complete scope-specific dependencies	Typed companion unavailable	Blocks only dependent claim; no implied zero

Both admitted diagnostics are signed:

$$D_{\text{RMS}} = \widehat{A}/\text{RMS}_{\text{full}}, \qquad Z_A = \widehat{A}/\sigma_A. \tag{4}$$

Neither is statistical significance. $|\widehat{A}|/\sigma_A$ has a distinct, unadmitted magnitude-standardization role; its exact identity is bound in the normative core. A legacy alias is admitted only after proof of mathematical identity and explicit method approval.

Scientifically important edge examples.

- Missing compatibility authority yields no numerical fit or fit-derived product.
- Unavailable source association blocks source-attributed displacement, amplitude, and intended-source diagnostics, not the numerical amplitude component.
- Circular shape preserves equal widths and invariant shape while orientation is symmetry-undefined.
- Missing formal-error authority can coexist with fitted values; marginal errors, unsupported joint covariance, and formal standardization remain unavailable.
- Unavailable response or bias is not zero and blocks only claims that depend on it.
- Failure of one array preserves conforming sibling atoms and never creates POINT whole-observation success or imputes the missing array.

6 Response families and displacement covariance

$$R^{\text{fixed}}_{\theta,m} = \left.\frac{\partial\widehat{\boldsymbol{\theta}}}{\partial m}\right|_{b,g,\,\text{support},W_{\text{fit}},\text{constraints}}, \tag{5}$$

$$R^{\text{fixed}}_{\theta,\text{source}} = R^{\text{fixed}}_{\theta,m} R^{\text{fixed}}_{m,\text{source}}, \tag{6}$$

$$R^{\text{POINT-FP}|\text{parent-fixed}}_{\theta,m}[\delta m] = \lim_{\epsilon\to 0} \frac{\widehat{\boldsymbol{\theta}}_{\mathcal{P}}(m+\epsilon\delta m) - \widehat{\boldsymbol{\theta}}_{\mathcal{P}}(m)}{\epsilon}. \tag{7}$$

Response family	What is held or rerun	Availability and non-equivalence
Fixed branch and fixed source composition	Branch, support, weights, constraints, and state fixed; compatible fixed parent response may compose	Requires exact local perturbation construction. It is neither full-procedure response nor empirical accuracy.
POINT full procedure, parent fixed	Rerun every POINT data-dependent branch while holding the realized parent product fixed	Requires comparable numerical outputs and separate authority. It is not fixed-branch or whole-chain response.
Whole-chain full procedure	Rerun every included upstream parent operation and POINT	Requires complete upstream-plus-POINT authority. It never aliases the parent-fixed response.

Full-procedure numerators subtract comparable numerical parameter outputs, not lifecycle or record objects. Comparability requires the same intended source, compatible association, identical roles, compatible gauge/order, units and basis, available components, and an exact perturbation-domain relation. Incomparable results are unavailable or typed discontinuities/state transitions. Finite differences are not multiplied as Jacobians without an exact composition theorem.

When every term is available in one common tangent basis, the displacement covariance is

$$C_\Delta = C_{\widehat{\boldsymbol{\mu}}} + C_{\boldsymbol{\mu}_{\mathrm{expected}}} - C_{\widehat{\boldsymbol{\mu}},\boldsymbol{\mu}_{\mathrm{expected}}} - C_{\boldsymbol{\mu}_{\mathrm{expected}},\widehat{\boldsymbol{\mu}}}. \tag{8}$$

No cross term is zero without independence authority. If any variance, cross-covariance, basis transform, or required Jacobian is unavailable, the combined covariance is unavailable while established component uncertainties remain visible. Shared expected-reference error does not average down as independent per-array noise without explicit authority.

7 Current status, evidence layers, and formal index

The generic conditional science has no unresolved owner decision. Remaining actions are content approvals: the three exact method records; exact numerical route instances; the four draft named-use profile identities and mechanics; and any separately requested response, bias/accuracy, or covariance authority. Only the POINT fit-completeness/publication decision participates in base fit completion. Pointing-support, telescope-QC, and CAL/TolProj decisions are immutable requested child artifacts. None may mutate the fit, create a new fit generation, rescue a sibling use, or block base completion merely because a downstream artifact is unrequested, unavailable, incomplete, failed, or absent.

Later evidence has distinct ceilings. Contract conformance tests identities, records, states, equations, dependencies, and failure behavior. Representation testing establishes faithful storage and transformation. Numerical testing can evaluate predictions only after relevant methods and routes exist. Observational validation separately measures bias, repeatability, coverage, or pointing performance. Production authorization also requires operational, performance, and deployment evidence. Passing one layer does not establish the next.

Normative requirements: SCI-POINT-REQ-001–038.
Falsifiable predictions: SCI-POINT-PRED-001–032.
Controlling source: SCI-POINT-NORMATIVE-CORE/v0.1-r0.4, exact SHA-256 `c0ca71bd457b8e6d37a425eb3ead76400dba3a5e29c869420807928201cdcdbd` shown on the cover.

This concise view claims no numerical route, implementation inspection or conformity, representation or response/covariance fidelity, uncertainty coverage, observational pointing accuracy, validation, performance, readiness, production suitability, production authorization, or Unity activity.